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Data Analyst

Portfolio Case Study



# 0.3Instacart Grocery Basket Analysis

## Company Overview

Instacart, an online grocery store that operates through an app. The task is to perform an initial data and exploratory analysis of some of their data in order to derive insights and suggest strategies for better segmentation based on some provided criteria

## Data

- The Instacart Online Grocery Shopping Dataset 2017”, Accessed from [www.instacart.com/datasets/grocery-shopping-2017](https://www.instacart.com/datasets/grocery-shopping-2017) via [Kaggle](#)

## Project Objective

Instacart already has very good sales, but they want to uncover more information about their sales patterns. They want to target different customers with applicable marketing campaigns

## Tools and Techniques Applied

- Annaconda/Jupyter Notebook
- Pandas
- Data wrangling & Subsetting
- Consistency checks
- Combining and Deriving
- Data Grouping & Aggregating
- Visualizations: created using matplotlib and Seaborn

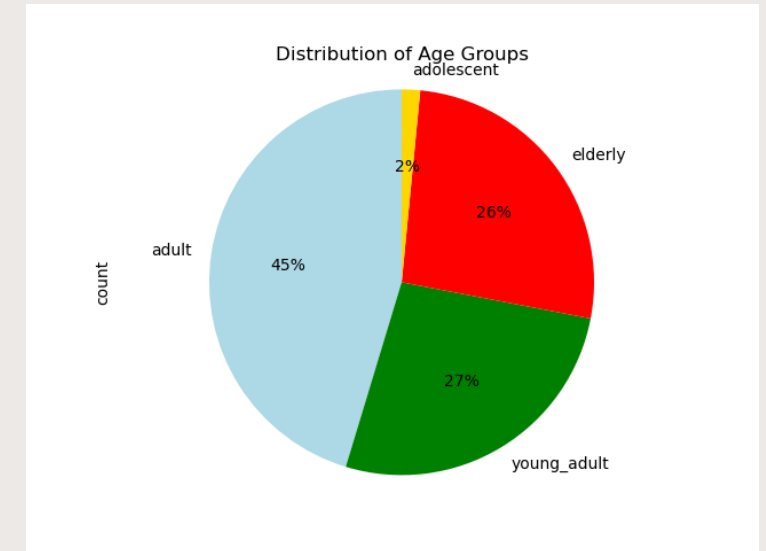
# Analysis

Started the project by cleaning and merging multiple datasets to ensure accuracy, using pandas for data wrangling, aggregation, and feature derivation.

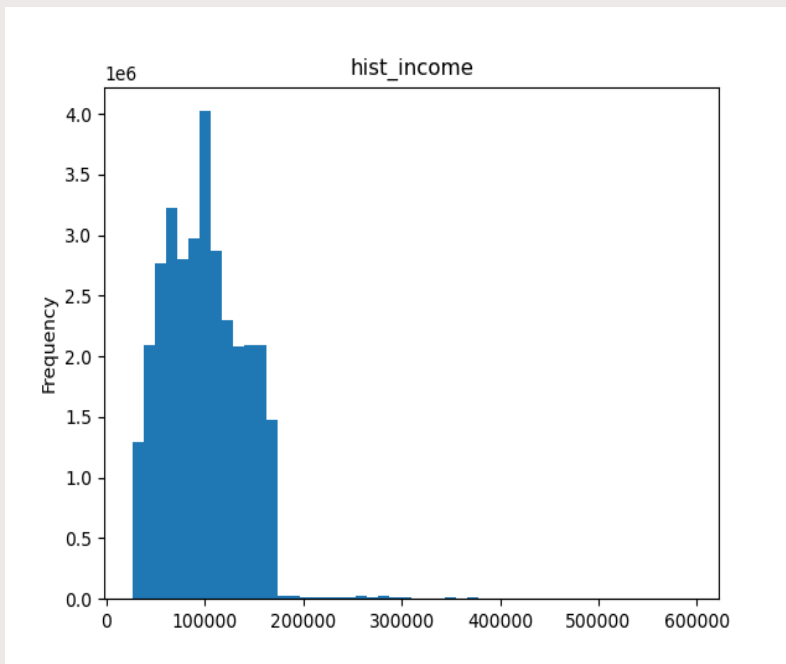
Identified key trends like peak shopping times and popular product categories, explored customer purchasing behaviours, and removed outliers to refine the dataset.

The comprehensive analysis provided actionable insights for crafting targeted marketing campaigns.

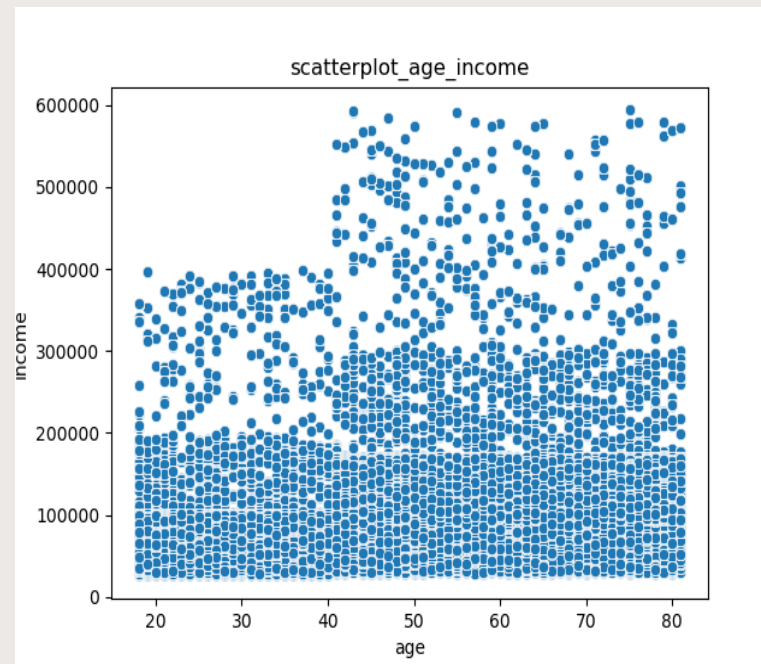
## Percentage of Age Groups



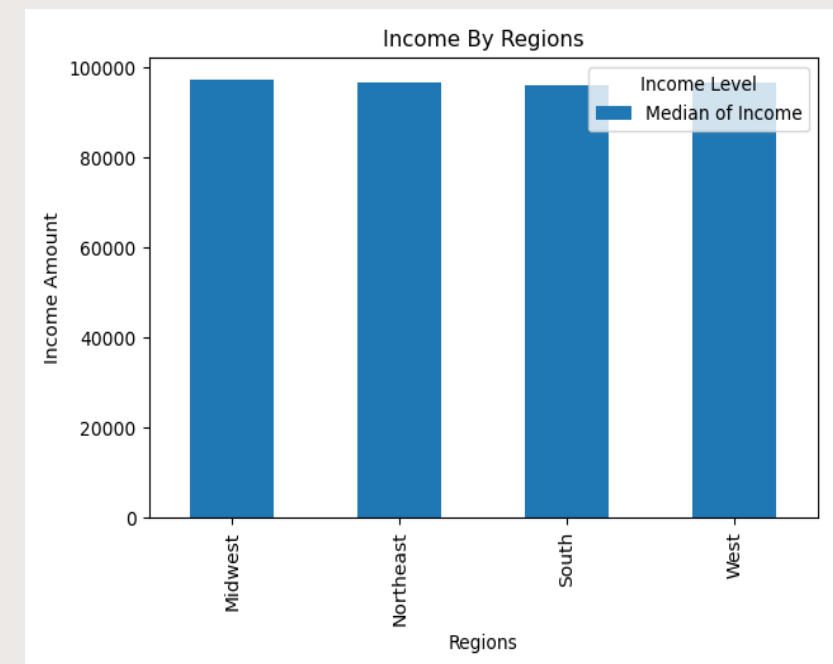
## Income Distribution of Customers



## Income in relation to Age



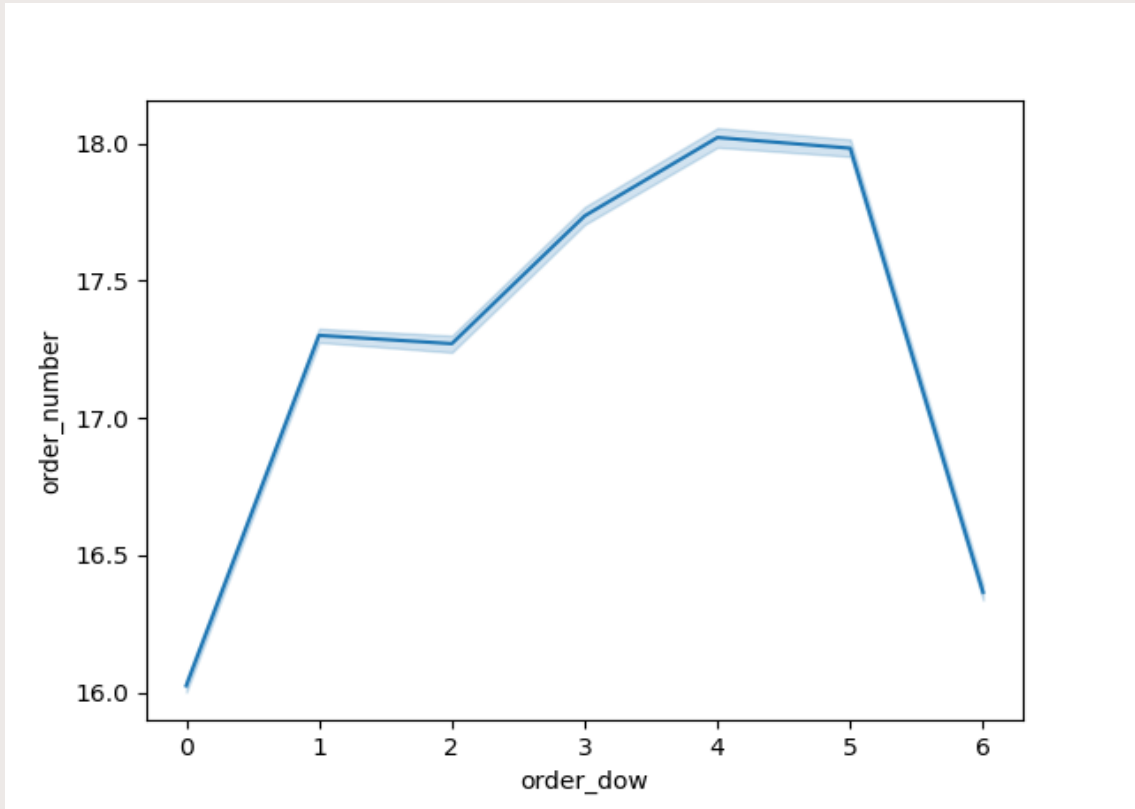
## Level of Income by Regions



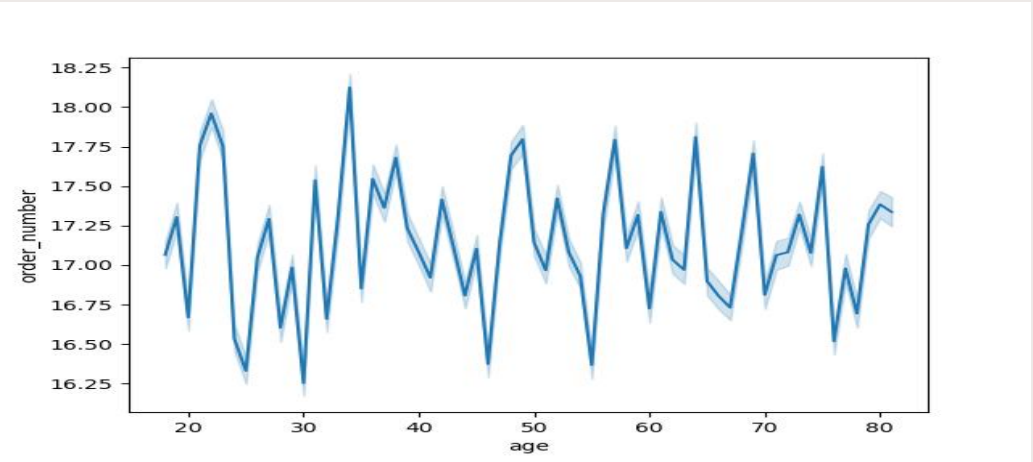
# Analysis

The analysis brought that the busiest ordering days are Saturdays and Sundays. Busiest time frame: 10am to 17pm. People tend to order to expensive items before 6-7pm. While most of the products people order are less than 10-15 USD.

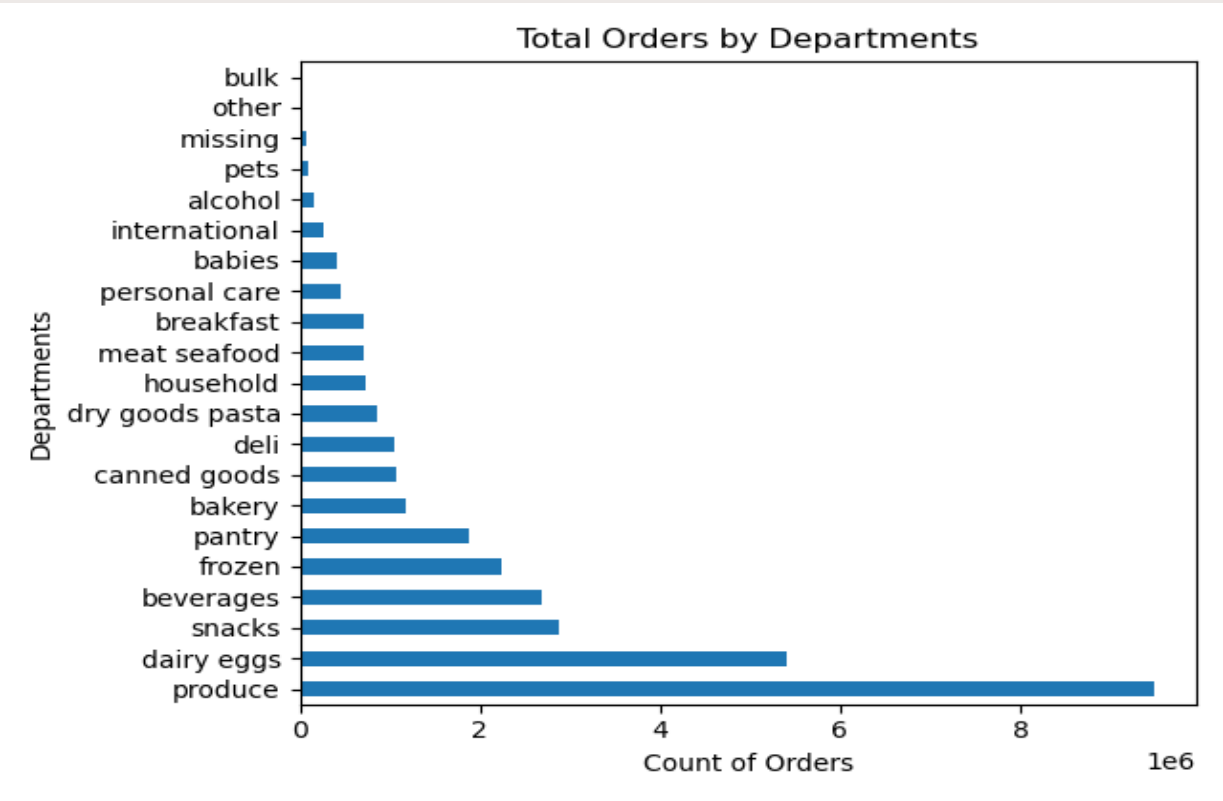
Orders Days of the Week



Age vs. Number of Order



Sales by Department



## KEY TAKEAWAYS

- People spend more on products normally on the weekends and during the core hours of the day. Advertisement can be promoted during these specific days and core hours .
- Products could be categorized as low-range, mid-range, and high-range to reflect price ranges. Overall, departments such as babies, dairy eggs, produce, snakes, and bakery show higher average sales. Instacart can redirect its marketing efforts to other products so as to increase their sales too.
- Instacart's customer base is mostly made up of regular customers (between 10 and 40 orders). It's not even different for the families of different size, meaning that larger families placed a similar number orders as smaller families. It could be beneficial to create a loyalty incentive (rewards, coupons, etc) to convert regular customers to loyal customers.
- The demographics show that younger families and individuals with children are more likely to purchase baby products, while older demographics may prioritize healthier options or specific categories like personal care. Customers with larger families tend to buy in bulk, particularly in the produce and dairy categories. Marketing promotions should be designed in the light of these realities.

# **CONTACT**

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**Github Repository**

<https://github.com/BASakhizada>